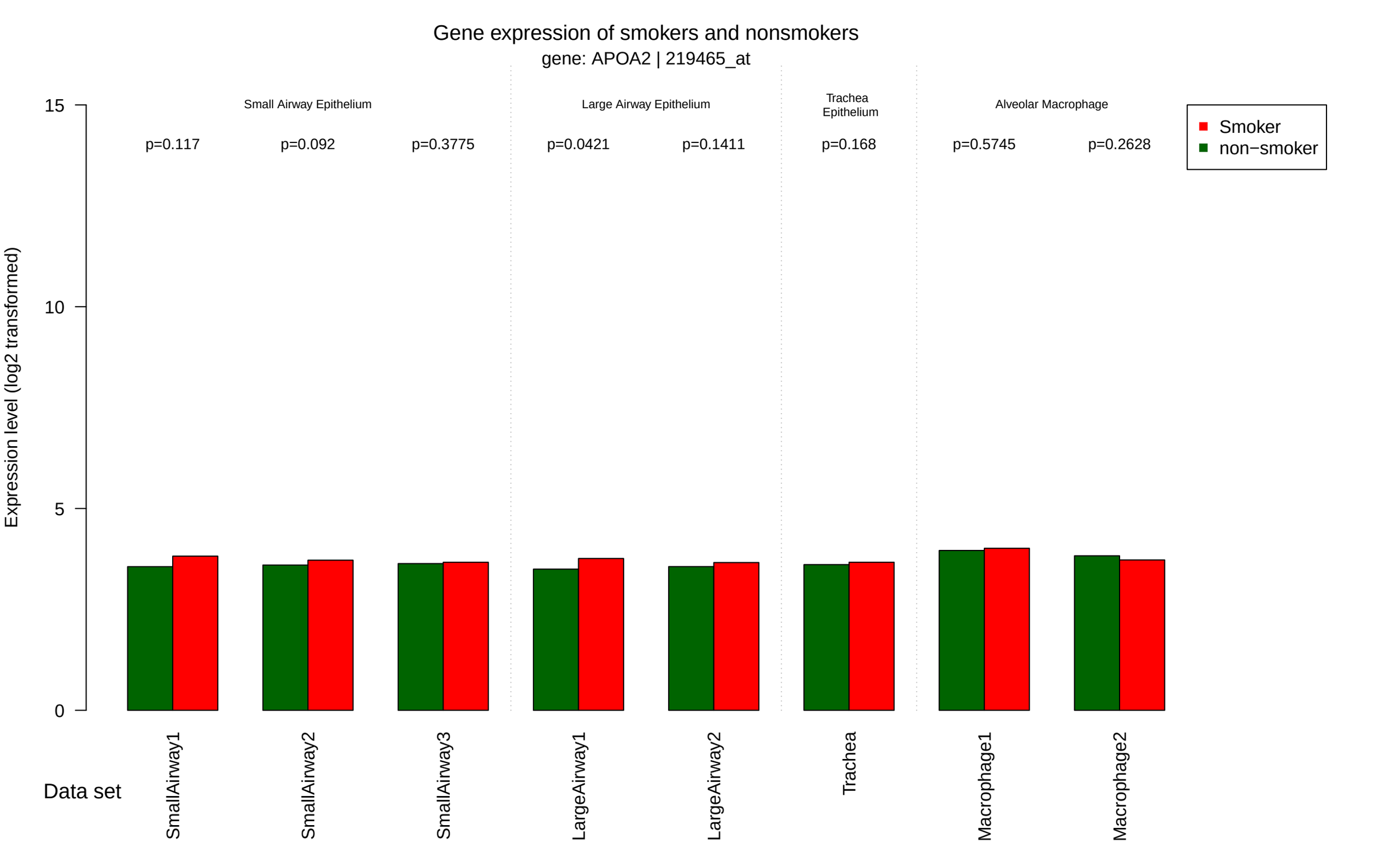
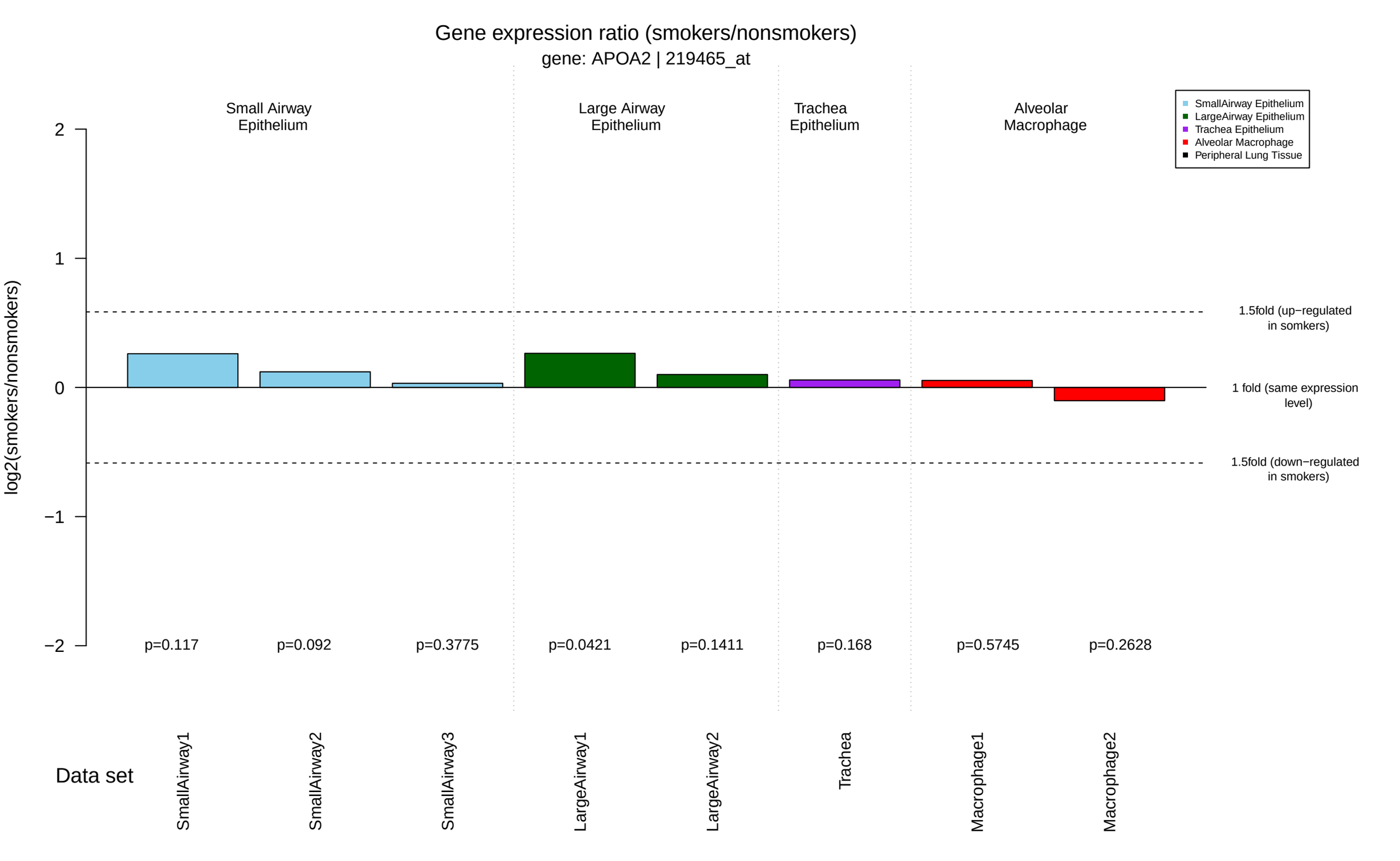
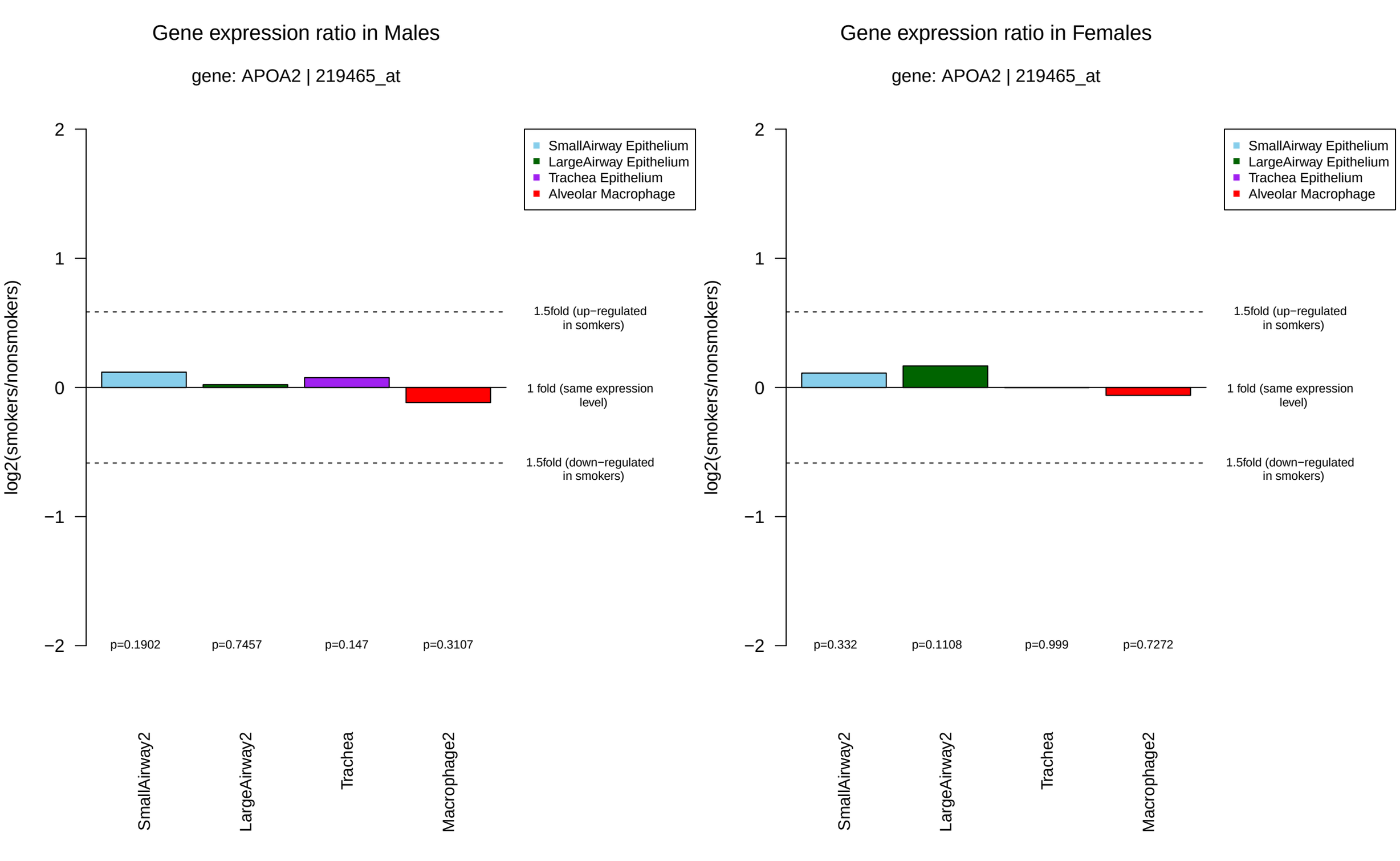
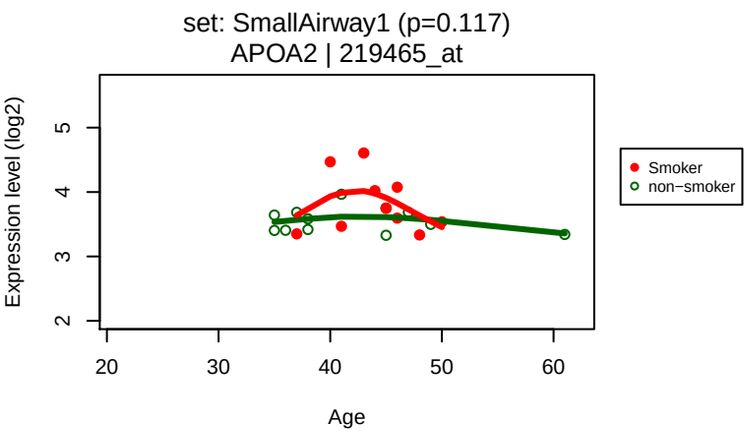
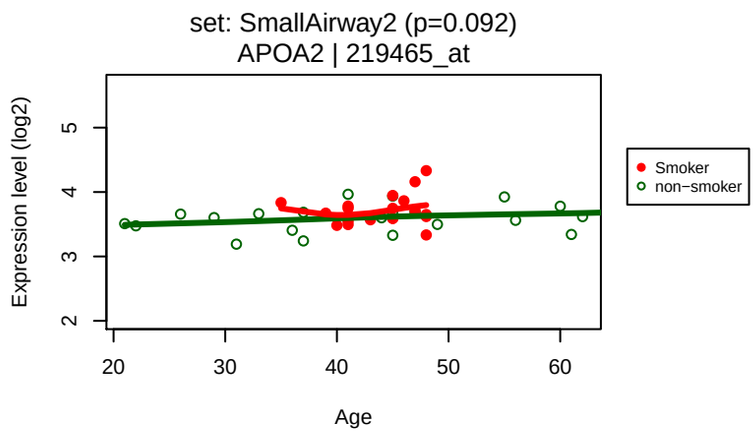
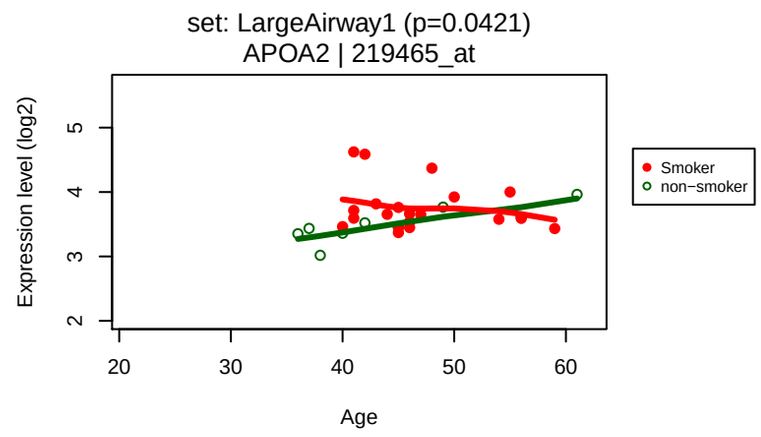

Gene expression of smokers and nonsmokers
gene: APOA2 | 219465_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.117
p=0.092
p=0.3775
p=0.0421
p=0.1411
p=0.168
p=0.5745
p=0.2628
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

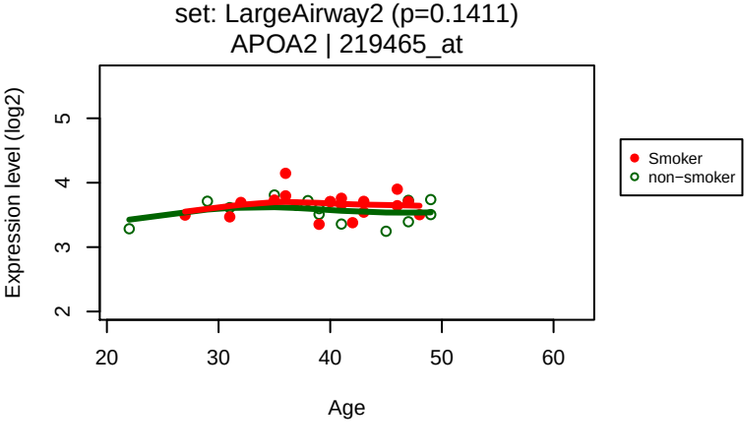

Gene expression ratio (smokers/nonsmokers)
gene: APOA2 | 219465_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.117
p=0.092
p=0.3775
p=0.0421
p=0.1411
p=0.168
p=0.5745
p=0.2628
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

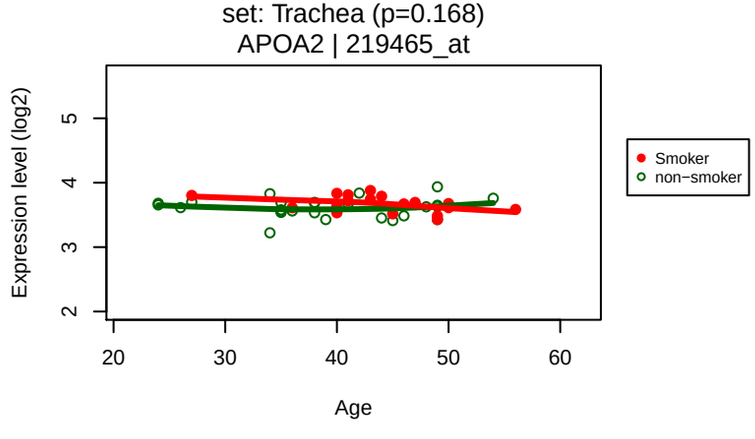

Gene expression ratio in Males
gene: APOA2 | 219465_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1902
p=0.7457
p=0.147
p=0.3107
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: APOA2 | 219465_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.332
p=0.1108
p=0.999
p=0.7272
SmallAirway2
LargeAirway2
Trachea
Macrophage2

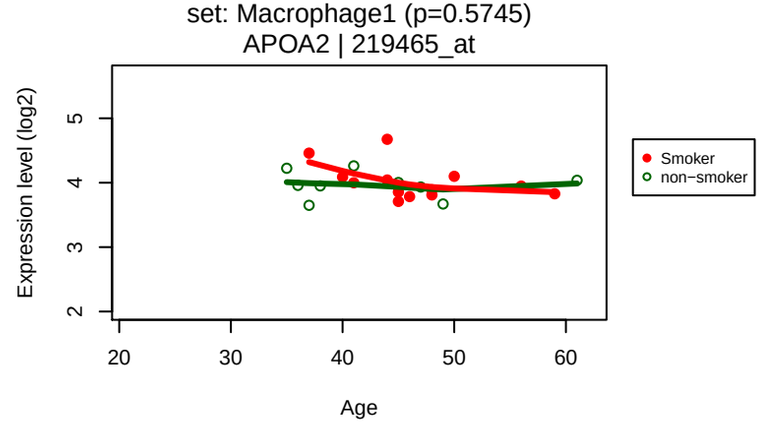

set: SmallAirway1 (p=0.117)
APOA2 | 219465_at
Expression level (log2)
Age
Smoker
non−smoker

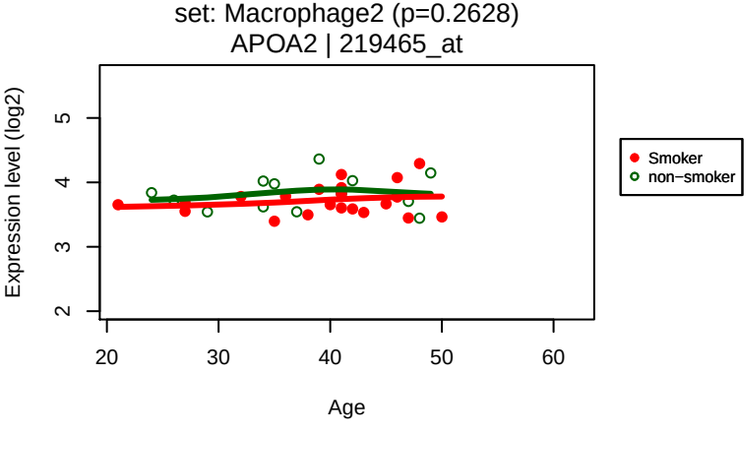

set: SmallAirway2 (p=0.092)
APOA2 | 219465_at
Expression level (log2)
Age
Smoker
non−smoker

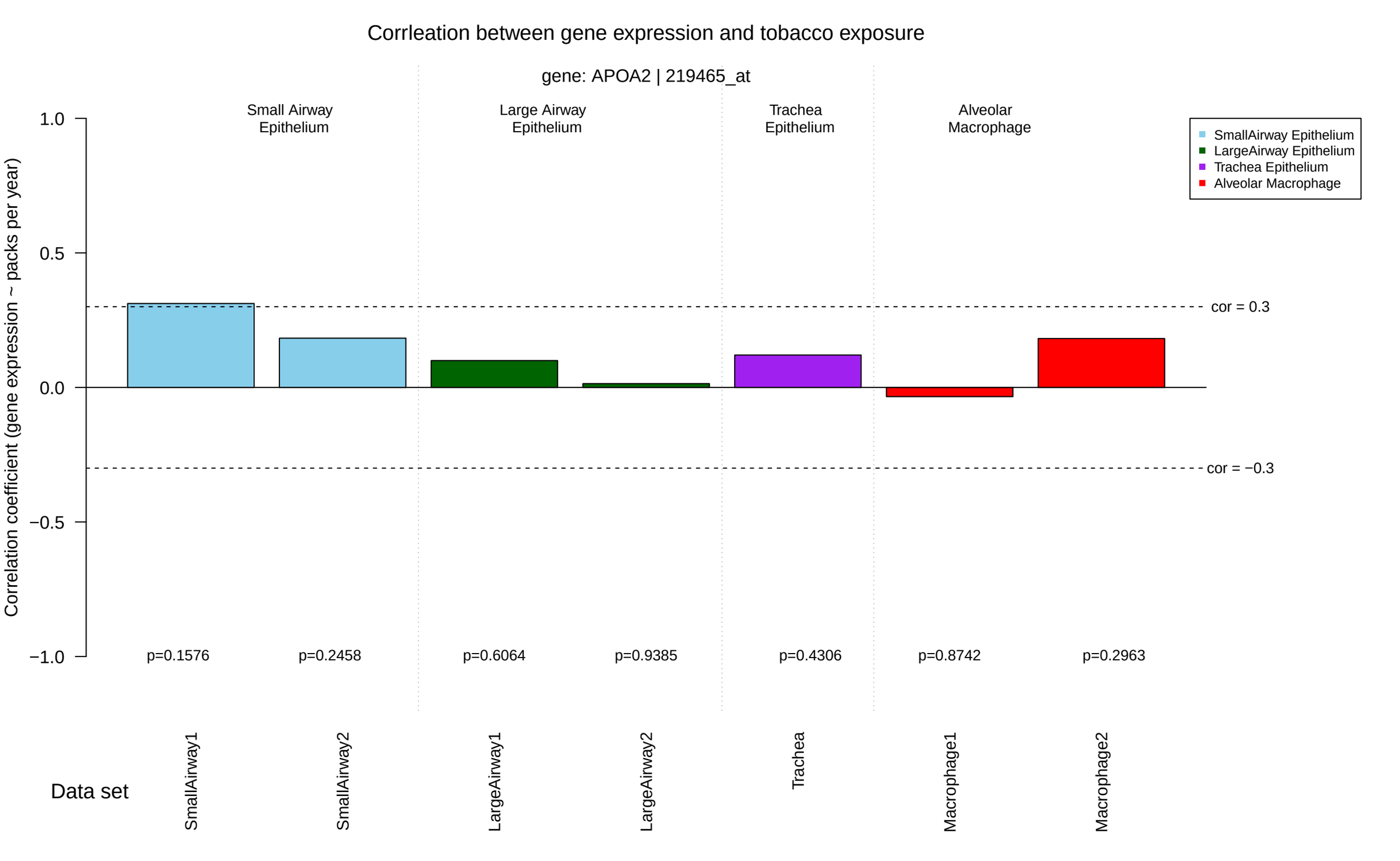

set: LargeAirway1 (p=0.0421)
APOA2 | 219465_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1411)
APOA2 | 219465_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.168)
APOA2 | 219465_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5745)
APOA2 | 219465_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2628)
APOA2 | 219465_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: APOA2 | 219465_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1576
p=0.2458
p=0.6064
p=0.9385
p=0.4306
p=0.8742
p=0.2963
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2